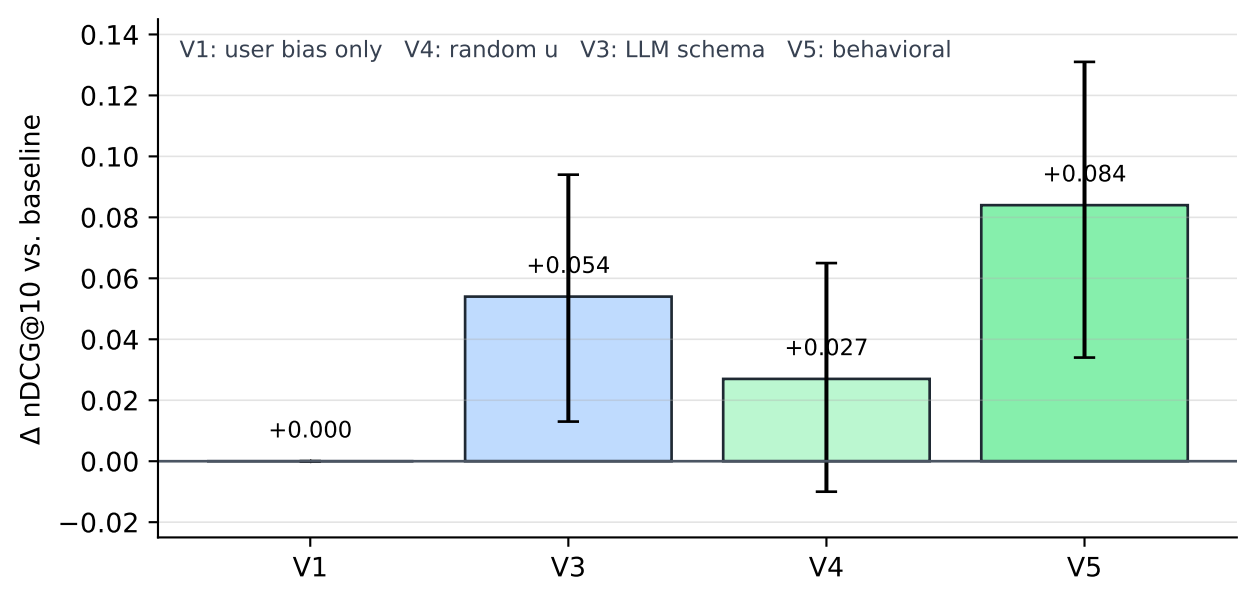
V1: user bias only
V4: random u
V3: LLM schema
V5: behavioral
Δ nDCG@10 vs. baseline
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
+0.000
+0.054
+0.027
+0.084
V1
V3
V4
V5